Aurora SVMAT Metric Formula Reference

Current V5 geometry and formula contract; V2/V3/V4 labels identify output families.

This document records the explicit mathematical definitions currently implemented in Aurora SVMAT Lab. The notation is aligned to the control-point sequence view c_i = {theta_i, w_i, M_i, z_i}.

RESEARCH USE ONLY. This document reflects the current code implementation and should be interpreted as a research reference rather than a clinical specification.

Notation

i indexes adjacent control-point intervals from control point i to i+1.

theta_i is the gantry angle in degrees at control point i.

z_i is the longitudinal or axial position in millimeters at control point i.

w_i is the cumulative meterset weight at control point i.

For all aperture-change fields, A_i is the physical double-layer intersection area after jaw clipping and the common 1 mm closure mask.

L_i is the total leaf-position vector at control point i across both layers.

delta theta_i = wrapped(theta_(i+1) - theta_i), constrained to [-180, 180] degrees.

delta z_i = z_(i+1) - z_i.

delta w_i = w_(i+1) - w_i.

delta A_i = |A_(i+1) - A_i|.

delta L_i = summed absolute leaf travel between adjacent control points across both layers.

CV(x) = std_pop(x) / mean(x), evaluated only on valid non-empty interval sets.

Section 1. V2 Metrics

longitudinal_travel_mm

Total absolute longitudinal travel

```
longitudinal_travel_mm = sum_i |delta z_i|
```

Meaning: Total absolute longitudinal travel reconstructed from adjacent control-point isocenter z positions.

total_rotation_deg

Total absolute gantry rotation

```
total_rotation_deg = sum_i |delta theta_i|
```

Meaning: Total absolute gantry rotation reconstructed from the observed angle sequence rather than declared direction tags.

rotations

Total gantry rotations

```
rotations = total_rotation_deg / 360
```

Meaning: Total gantry rotation expressed in full 360-degree turns.

travel_per_rotation_mm

Longitudinal travel per full gantry rotation

```
travel_per_rotation_mm = longitudinal_travel_mm / rotations
```

Meaning: Average longitudinal travel per gantry rotation, derived from total travel and total rotation.

projection_pitch_mean

Mean projection pitch

```
projection_pitch_i = |delta z_i| / |delta theta_i|
projection_pitch_mean = mean_i(projection_pitch_i)
```

Meaning: Mean projection pitch, defined as the mean of |delta z| / |delta theta| over valid adjacent control-point intervals.

projection_pitch_cv

Projection pitch variability

```
projection_pitch_i = |delta z_i| / |delta theta_i|
projection_pitch_cv = CV(projection_pitch_i)
```

Meaning: Coefficient of variation of projection pitch. Lower values mean steadier theta-z coupling.

projection_mu_density_mean_proxy

Mean projection meterset-weight density proxy

```
projection_mu_density_proxy_i = delta w_i / |delta z_i|
projection_mu_density_mean_proxy = mean_i(projection_mu_density_proxy_i)
```

Meaning: Mean projection MU-density proxy, using cumulative meterset-weight change per millimeter of longitudinal travel.

projection_mu_density_cv_proxy

Variability of the projection meterset-weight density proxy

```
projection_mu_density_proxy_i = delta w_i / |delta z_i|
projection_mu_density_cv_proxy = CV(projection_mu_density_proxy_i)
```

Meaning: Coefficient of variation of the projection MU-density proxy across adjacent control-point intervals.

projection_aperture_change_mean

Mean aperture-change rate along the longitudinal axis

```
delta A_i = |A_(i+1) - A_i|
projection_aperture_change_i = delta A_i / |delta z_i|
projection_aperture_change_mean = mean_i(projection_aperture_change_i)
```

Meaning: Mean absolute change in physical dual-layer intersection area per millimeter of couch travel, mm^2/mm; the 1 mm closure mask is shared with SAS and MCS.

projection_aperture_change_cv

Variability of the aperture-change rate

```
delta A_i = |A_(i+1) - A_i|
projection_aperture_change_i = delta A_i / |delta z_i|
projection_aperture_change_cv = CV(projection_aperture_change_i)
```

Meaning: Coefficient of variation of the projection aperture-change rate.

projection_leaf_travel_mean

Mean leaf-travel rate across both MLC layers

```
delta L_i = sum_leaves |leaf_(i+1) - leaf_i| across MLCX1 and MLCX2
projection_leaf_travel_i = delta L_i / |delta z_i|
projection_leaf_travel_mean = mean_i(projection_leaf_travel_i)
```

Meaning: Mean projection leaf-travel rate, defined as summed leaf travel per millimeter of longitudinal travel.

projection_leaf_travel_cv

Variability of the total leaf-travel rate

```
delta L_i = sum_leaves |leaf_(i+1) - leaf_i| across MLCX1 and MLCX2
projection_leaf_travel_i = delta L_i / |delta z_i|
projection_leaf_travel_cv = CV(projection_leaf_travel_i)
```

Meaning: Coefficient of variation of the projection leaf-travel rate.

projection_leaf_travel_mean_mlcx1

Mean leaf-travel rate for MLCX1

```
delta L_i^(MLCX1) = sum_x1 |x1_(i+1) - x1_i|
projection_leaf_travel_mlcx1_i = delta L_i^(MLCX1) / |delta z_i|
projection_leaf_travel_mean_mlcx1 = mean_i(projection_leaf_travel_mlcx1_i)
```

Meaning: Mean projection leaf-travel rate for the MLCX1 layer only, normalized by longitudinal travel.

projection_leaf_travel_cv_mlcx1

Variability of the MLCX1 leaf-travel rate

```
delta L_i^(MLCX1) = sum_x1 |x1_(i+1) - x1_i|
projection_leaf_travel_mlcx1_i = delta L_i^(MLCX1) / |delta z_i|
projection_leaf_travel_cv_mlcx1 = CV(projection_leaf_travel_mlcx1_i)
```

Meaning: Coefficient of variation of the MLCX1 projection leaf-travel rate.

projection_leaf_travel_mean_mlcx2

Mean leaf-travel rate for MLCX2

```
delta L_i^(MLCX2) = sum_x2 |x2_(i+1) - x2_i|
projection_leaf_travel_mlcx2_i = delta L_i^(MLCX2) / |delta z_i|
projection_leaf_travel_mean_mlcx2 = mean_i(projection_leaf_travel_mlcx2_i)
```

Meaning: Mean projection leaf-travel rate for the MLCX2 layer only, normalized by longitudinal travel.

projection_leaf_travel_cv_mlcx2

Variability of the MLCX2 leaf-travel rate

```
delta L_i^(MLCX2) = sum_x2 |x2_(i+1) - x2_i|
projection_leaf_travel_mlcx2_i = delta L_i^(MLCX2) / |delta z_i|
projection_leaf_travel_cv_mlcx2 = CV(projection_leaf_travel_mlcx2_i)
```

Meaning: Coefficient of variation of the MLCX2 projection leaf-travel rate.

theta_z_coupling_cv

Theta-z coupling variability

```
theta_z_coupling_cv = projection_pitch_cv
```

Meaning: Theta-z coupling variability, reported here as the same projection-pitch coefficient of variation.

mu_z_coupling_cv_proxy

Meterset-weight-density to z coupling variability proxy

```
mu_z_coupling_cv_proxy = projection_mu_density_cv_proxy
```

Meaning: MU-z coupling variability proxy, reported as the coefficient of variation of meterset-weight density over longitudinal travel.

mlc_z_coupling_cv

MLC-z coupling variability across both layers

```
mlc_z_coupling_cv = projection_leaf_travel_cv
```

Meaning: MLC-z coupling variability, reported here as the coefficient of variation of the projection leaf-travel rate.

mlcx1_z_coupling_cv

MLCX1-z coupling variability

```
mlcx1_z_coupling_cv = projection_leaf_travel_cv_mlcx1
```

Meaning: MLCX1-z coupling variability, reported as the coefficient of variation of the MLCX1 projection leaf-travel rate.

mlcx2_z_coupling_cv

MLCX2-z coupling variability

```
mlcx2_z_coupling_cv = projection_leaf_travel_cv_mlcx2
```

Meaning: MLCX2-z coupling variability, reported as the coefficient of variation of the MLCX2 projection leaf-travel rate.

Section 2. V3 Research Metrics

These research-facing v3 metrics are the immediately implementable Aurora extensions derived directly from currently parsed RTPLAN control-point data.

reversal_symmetry_index

Bidirectional modulation symmetry

```
mean over selected burden families of [1 - abs(forward_mean - backward_mean) /
(abs(forward_mean) + abs(backward_mean))]
```

Meaning: Symmetry score comparing forward and backward beam groups across matched modulation summaries. Higher values mean more balanced bidirectional modulation.

forward_backward_metric_difference

Forward-versus-backward modulation difference

```
abs(mean_forward(coupled_modulation_index) - mean_backward(coupled_modulation_index))
```

Meaning: Absolute difference in average modulation burden between forward-moving and backward-moving beam groups.

beam_pair_balance_index

Beam-pair modulation balance

```
1 - abs(sum_forward(coupled_modulation_index) - sum_backward(coupled_modulation_index)) /
(sum_forward + sum_backward)
```

Meaning: Balance score comparing the total modulation burden assigned to forward and backward beam groups.

small_opening_fraction

Fraction of narrow effective openings

```
mean_sas10; OpenMUmean(count_s(1+1e-8<g<10 mm)/count_s(g>1+1e-8))
```

Meaning: Open-aperture delivered-MU mean of the fraction of effective sub-strips with 1+1e-8 < gap < 10 mm; exactly the same value as mean_sas10.

near_closed_fraction

Fraction of near-closed effective openings

```
OpenMUmean(count_s(1+1e-8<g<2 mm)/count_s(g>1+1e-8))
```

Meaning: Open-aperture delivered-MU mean fraction of effective sub-strips with 1+1e-8 < gap < 2 mm; the same closure mask and denominator as SAS.

effective_small_gap_burden

Weighted burden of narrow effective gaps

```
OpenMUmean(mean_open_strips(max(0,1-g/10 mm))); same 1 mm closure and delivered-MU weights
as SAS
```

Meaning: Open-aperture delivered-MU mean of the within-CP average max(0,1-gap/10 mm) over open effective sub-strips; no zero-MU fallback.

projection_pitch_p95

95th percentile of projection pitch

```
nearest-rank 95th percentile of projection_pitch_i
```

Meaning: 95th percentile of projection pitch, using the nearest-rank interval summary.

projection_pitch_max

Maximum projection pitch

```
max(projection_pitch_i)
```

Meaning: Maximum projection pitch observed across adjacent control-point intervals.

projection_pitch_top3_mean

Mean of the top three projection-pitch intervals

```
mean of the three largest projection_pitch_i values
```

Meaning: Mean of the three largest projection-pitch intervals, or all intervals if fewer than three exist.

projection_mu_density_p95_proxy

95th percentile of the projection meterset-weight density proxy

```
nearest-rank 95th percentile of projection_mu_density_proxy_i
```

Meaning: 95th percentile of the projection MU-density proxy, using cumulative meterset-weight change per millimeter.

projection_mu_density_max_proxy

Maximum projection meterset-weight density proxy

```
max(projection_mu_density_proxy_i)
```

Meaning: Maximum projection MU-density proxy observed across adjacent control-point intervals.

projection_mu_density_top3_mean_proxy

Mean of the top three projection meterset-weight density proxy values

```
mean of the three largest projection_mu_density_proxy_i values
```

Meaning: Mean of the three largest projection MU-density proxy values.

projection_aperture_change_p95

95th percentile of the aperture-change rate

```
nearest-rank 95th percentile of projection_aperture_change_i
```

Meaning: 95th percentile of the projection aperture-change rate.

projection_aperture_change_max

Maximum aperture-change rate

```
max(projection_aperture_change_i)
```

Meaning: Maximum projection aperture-change rate observed across adjacent control-point intervals.

projection_aperture_change_top3_mean

Mean of the top three aperture-change rates

```
mean of the three largest projection_aperture_change_i values
```

Meaning: Mean of the three largest projection aperture-change rates.

projection_leaf_travel_p95

95th percentile of the leaf-travel rate

```
nearest-rank 95th percentile of projection_leaf_travel_i
```

Meaning: 95th percentile of the projection leaf-travel rate across both layers.

projection_leaf_travel_max

Maximum leaf-travel rate

```
max(projection_leaf_travel_i)
```

Meaning: Maximum projection leaf-travel rate across both layers.

projection_leaf_travel_top3_mean

Mean of the top three leaf-travel rates

```
mean of the three largest projection_leaf_travel_i values
```

Meaning: Mean of the three largest projection leaf-travel rates across both layers.

head_mu_density_mean_proxy

Head-third mean MU-density proxy

```
mean(projection_mu_density_proxy_i) over head-third intervals
```

Meaning: Mean projection MU-density proxy inside the head longitudinal third of the covered z-range.

head_mu_density_cv_proxy

Head-third MU-density proxy variability

```
CV(projection_mu_density_proxy_i) over head-third intervals
```

Meaning: Coefficient of variation of the projection MU-density proxy inside the head longitudinal third.

head_aperture_change_mean

Head-third mean aperture-change rate

```
mean(projection_aperture_change_i) over head-third intervals
```

Meaning: Mean projection aperture-change rate inside the head longitudinal third of the covered z-range.

head_aperture_change_cv

Head-third aperture-change variability

```
CV(projection_aperture_change_i) over head-third intervals
```

Meaning: Coefficient of variation of the projection aperture-change rate inside the head longitudinal third.

head_leaf_travel_mean

Head-third mean leaf-travel rate

```
mean(projection_leaf_travel_i) over head-third intervals
```

Meaning: Mean projection leaf-travel rate inside the head longitudinal third of the covered z-range.

head_leaf_travel_cv

Head-third leaf-travel variability

```
CV(projection_leaf_travel_i) over head-third intervals
```

Meaning: Coefficient of variation of the projection leaf-travel rate inside the head longitudinal third.

mid_mu_density_mean_proxy

Mid-third mean MU-density proxy

```
mean(projection_mu_density_proxy_i) over mid-third intervals
```

Meaning: Mean projection MU-density proxy inside the mid longitudinal third of the covered z-range.

mid_mu_density_cv_proxy

Mid-third MU-density proxy variability

```
CV(projection_mu_density_proxy_i) over mid-third intervals
```

Meaning: Coefficient of variation of the projection MU-density proxy inside the mid longitudinal third.

mid_aperture_change_mean

Mid-third mean aperture-change rate

```
mean(projection_aperture_change_i) over mid-third intervals
```

Meaning: Mean projection aperture-change rate inside the mid longitudinal third of the covered z-range.

mid_aperture_change_cv

Mid-third aperture-change variability

```
CV(projection_aperture_change_i) over mid-third intervals
```

Meaning: Coefficient of variation of the projection aperture-change rate inside the mid longitudinal third.

mid_leaf_travel_mean

Mid-third mean leaf-travel rate

```
mean(projection_leaf_travel_i) over mid-third intervals
```

Meaning: Mean projection leaf-travel rate inside the mid longitudinal third of the covered z-range.

mid_leaf_travel_cv

Mid-third leaf-travel variability

```
CV(projection_leaf_travel_i) over mid-third intervals
```

Meaning: Coefficient of variation of the projection leaf-travel rate inside the mid longitudinal third.

tail_mu_density_mean_proxy

Tail-third mean MU-density proxy

```
mean(projection_mu_density_proxy_i) over tail-third intervals
```

Meaning: Mean projection MU-density proxy inside the tail longitudinal third of the covered z-range.

tail_mu_density_cv_proxy

Tail-third MU-density proxy variability

```
CV(projection_mu_density_proxy_i) over tail-third intervals
```

Meaning: Coefficient of variation of the projection MU-density proxy inside the tail longitudinal third.

tail_aperture_change_mean

Tail-third mean aperture-change rate

```
mean(projection_aperture_change_i) over tail-third intervals
```

Meaning: Mean projection aperture-change rate inside the tail longitudinal third of the covered z-range.

tail_aperture_change_cv

Tail-third aperture-change variability

```
CV(projection_aperture_change_i) over tail-third intervals
```

Meaning: Coefficient of variation of the projection aperture-change rate inside the tail longitudinal third.

tail_leaf_travel_mean

Tail-third mean leaf-travel rate

```
mean(projection_leaf_travel_i) over tail-third intervals
```

Meaning: Mean projection leaf-travel rate inside the tail longitudinal third of the covered z-range.

tail_leaf_travel_cv

Tail-third leaf-travel variability

```
CV(projection_leaf_travel_i) over tail-third intervals
```

Meaning: Coefficient of variation of the projection leaf-travel rate inside the tail longitudinal third.

layer_imbalance_index

Dual-layer travel imbalance

```
abs(mean(leaf_travel_i_MLCX1) - mean(leaf_travel_i_MLCX2)) / (abs(mean_MLCX1) +
abs(mean_MLCX2))
```

Meaning: Normalized difference between the mean MLCX1 and MLCX2 leaf-travel rates.

layer_correlation_index

Dual-layer travel correlation

```
correlation(projection_leaf_travel_i_MLCX1, projection_leaf_travel_i_MLCX2)
```

Meaning: Correlation between the interval-by-interval MLCX1 and MLCX2 leaf-travel rates.

x1_x2_aperture_disparity

Physical dual-layer area disparity

```
OpenMUmean(abs(A_MLCX1-A_MLCX2)/(A_MLCX1+A_MLCX2)); native jaw-clipped areas on common Y
coverage with 1 mm closure
```

Meaning: Open-aperture delivered-MU mean abs(A1-A2)/(A1+A2); native physical layer areas use the same Y coverage, jaws and 1 mm closure mask.

x1_x2_leaf_travel_ratio

Dual-layer leaf-travel ratio

```
mean(projection_leaf_travel_i_MLCX1) / mean(projection_leaf_travel_i_MLCX2)
```

Meaning: Ratio of mean MLCX1 leaf-travel rate to mean MLCX2 leaf-travel rate.

Section 3. Unified Physical Aperture Metrics

Version: aurora-v5-unified-physical-closure1mm. Real leaf boundaries define a common y-sub-strip grid. Each layer's position vector is split into A/B banks; intervals are intersected across layers and clipped to jaws. A and P are exact area and union perimeter. These seven descriptors are a physical-geometry extension with adapted MCS, not exact McNiven MCS or a reproduction of the old unit-width design. Effective gaps <=1+1e-8 mm are closed. Only positive cumulative-weight increments contribute, represented by endpoint i+1. Weights are delta CMW / beam final CMW times BeamMeterset. If any BeamMeterset is absent, every beam uses unit-normalized relative weights and a warning. Missing geometry or malformed/decreasing weights makes all seven unavailable. All seven use open-aperture conditional means: exclude area-zero endpoints and renormalize over retained weights. Closed leaves do not enter SAS or LSV. Exclusion count and weight fraction are disclosed in warnings; no open endpoint makes all seven unavailable. Endpoint sampling does not imply the whole interval was closed.

mean_ba

mean_ba

```
OpenMUmean(A_i); A_i=sum_s h_i,s*g_i,s, exact physical dual-layer intersection area
```

Meaning: Open-aperture conditional delivered-weight mean physical area with the 1 mm closure mask after intersecting both MLC layers and clipping to X/Y jaws using real leaf boundaries; closed endpoints excluded and weights renormalized.

mean_bi

mean_bi

```
OpenMUmean(P_i^2/(4*pi*A_i)); P_i is the exact aperture-union perimeter; A_i=0 => excluded
```

Meaning: Open-aperture conditional delivered-weight mean P squared / (4 pi A), using exact physical union perimeter; closed endpoints excluded and weights renormalized; unavailable only when no valid open aperture remains.

mean_ca

mean_ca

```
OpenMUmean(P_i/A_i); units mm^-1; A_i=0 => excluded; this is not circularity
```

Meaning: Open-aperture conditional delivered-weight mean P/A in inverse millimeters, not circularity; closed endpoints excluded and weights renormalized.

mean_sas5

mean_sas5

```
OpenMUmean(count_s(1+1e-8<g_i,s<5 mm)/count_s(g_i,s>1+1e-8)); positive-height sub-strips
only; closed => excluded
```

Meaning: Open-aperture conditional delivered-weight mean fraction of open physical sub-strips (gap > 1+1e-8 mm) with gap strictly below 5 mm; equal sub-strip counts; closed leaves and fully closed endpoints excluded.

mean_sas10

mean_sas10

```
OpenMUmean(count_s(1+1e-8<g_i,s<10 mm)/count_s(g_i,s>1+1e-8)); positive-height sub-strips
only; closed => excluded
```

Meaning: Open-aperture conditional delivered-weight mean fraction of open physical sub-strips (gap > 1+1e-8 mm) with gap strictly below 10 mm; equal sub-strip counts; closed leaves and fully closed endpoints excluded.

mean_mcs_aurora

mean_mcs_aurora

```
OpenMUmean(AAV_i*LSV_i); AAV_i=A_i/sum_s max_positive_endpoint(h_i,s*g_i,s);
LSV_i=1-sum_adj_open abs(delta g)/((n_open-1)*max_open(g)); n_open<=1 => LSV 1; zero
envelope => MCS 0; including moving Y-jaw clipping
```

Meaning: Open-aperture conditional delivered-weight mean AAV times open-gap LSV (gap > 1+1e-8 mm). AAV uses beam maxima of clipped sub-strip areas. Closed endpoints excluded; weights renormalized. Adapted MCS, not exact McNiven MCS.

mcs_complexity_aurora

mcs_complexity_aurora

```
1-mean_mcs_aurora; complement of the physical-aperture adapted MCS
```

Meaning: One minus the Aurora physical-aperture adapted MCS; larger values indicate greater modulation complexity under this version-specific definition.

Appendix A. Legacy Engineering Metrics

These legacy metrics are retained for engineering comparison with the first-pass Aurora prototype. They are not the preferred paper-style reporting set.

axial_travel_mm

Legacy total axial travel

```
axial_travel_mm = sum_i |delta z_i|
```

Meaning: Legacy engineering metric for total absolute axial travel reconstructed from adjacent control-point isocenter z positions.

gantry_rotation_deg

Legacy total gantry rotation

```
gantry_rotation_deg = sum_i |delta theta_i|
```

Meaning: Legacy engineering metric for total absolute gantry rotation reconstructed from the observed angle sequence.

mm_per_deg

Legacy axial travel per gantry degree

```
mm_per_deg = axial_travel_mm / gantry_rotation_deg
```

Meaning: Legacy engineering metric for total axial travel divided by total gantry rotation.

mm_per_rotation

Legacy axial travel per full rotation

```
mm_per_rotation = axial_travel_mm / (gantry_rotation_deg / 360)
```

Meaning: Legacy engineering metric for total axial travel per full 360-degree gantry rotation.

pitch_consistency

Legacy pitch consistency metric

```
interval_pitch_i = |delta z_i| / |delta theta_i|
pitch_consistency = CV(interval_pitch_i)
```

Meaning: Legacy engineering pitch variability metric, equivalent to the coefficient of variation of interval pitch.

mu_per_mm

Legacy MU density along the axial axis

```
mu_per_mm = sum_absolute_BeamMeterset_MU / axial_travel_mm; missing MU => unavailable
```

Meaning: Absolute DICOM BeamMeterset MU summed across beams, divided by total axial travel. Missing or invalid beam MU yields unavailable, never a relative-weight proxy.

aperture_change_per_mm

Legacy aperture-change density

```
aperture_change_per_mm = sum_i delta A_i / sum_i |delta z_i|
```

Meaning: Total absolute effective area change divided by total positive couch travel, mm^2/mm; physical layers are intersected on their boundary grid and clipped to jaws.

leaf_travel_per_mm

Legacy leaf-travel density

```
leaf_travel_per_mm = sum_i delta L_i / sum_i |delta z_i|
```

Meaning: Legacy engineering metric for total absolute leaf travel normalized by total axial travel.

coupled_modulation_index

Legacy engineering aggregate

```
n(x) = x / (1 + x)
coupled_modulation_index = mean( n(aperture_change_per_mm), n(leaf_travel_per_mm),
n(mu_density_variability), n(pitch_consistency) )
```

Meaning: Equal mean of x/(1+x) for area-change rate scaled by 1 mm^2/mm, leaf-travel rate scaled by 1 mm/mm, and the two CVs. Required physical area geometry must be valid.